Eli R. Hooten, Ph.D.

Curriculum Vitae

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EXECUTIVE SUMMARY

Since graduating in 2014 with my Ph.D. in Computer Science, I developed a deep knowledge of experimental design, human factors, statistical analysis, software development, and team management. I have applied these skills in technical leadership positions across several startups.

As the Co-Founder/CTO of GameWisp I grew the software team to ten individuals comprised of developers, designers, data scientists, and project managers. I led the overall technical design of GameWisp's products and shipped production-quality code on a daily basis, and grew our Annual Recurring Revenue to over $5M.

While at Codecov I served as Co-Founder/CTO and scaled the technical team from 3 to 30+ across diverse roles. I also led the overall technical direction that turned Codecov into a world class, industry leading developer tool. My efforts grew Codecov's revenue and user base many times over, and as of 2022 Codecov has been used by over a million software developers worldwide. I also conducted peer-reviewed and published research on code coverage and its usage in real engineering teams. Codecov was acquired by Sentry in 2022, where I served as Director of Engineering until 2025.

I am currently Co-Founder/CTO of Line, where I build an AI sales agent that works inbound event inquiries for hospitality venues. The agent is live at venues across the US and Canada, where it resolves more than 80% of the conversations it handles with no human involvement. It is grounded by systems I designed to verify every factual claim it makes before a guest sees it. I also built the AI-first engineering pipeline behind the product, which let a two-engineer team land over 1,600 high quality, fully reviewed pull requests in fourteen months.

I enjoy applying my skills as an executive leader, software developer, and scientist to technology companies of all sizes. I'm passionate about managing product and engineering teams, solving complex technical problems, and providing executive-level oversight to software businesses.

CITIZENSHIP

USA

TECHNICAL EXPERTISE

Executive leadership; Management; Leading product and engineering teams; Designing, scoping, and building high scale web-based applications; Organizational process development and implementation.

EDUCATION

Vanderbilt University, Nashville, TN USA

Ph.D., Computer Science, May 2014

- Dissertation: Information Context in Geocollaborative Interfaces
- Advisor: Professor Julie A. Adams
- Areas of Study: Data Visualization, Human Factors

M.Sc., Computer Science, May 2013

- Advisor: Professor Julie A. Adams
- GPA: 3.91/4.0

M.Sc., Electrical Engineering, December 2010

- Thesis: A Mobile, Map-Based Tasking Interface for Human-Robot Interaction
- Advisor: Professor Julie A. Adams
- Area of Study: Human-Robot Interaction
- GPA: 3.86/4.0

Murray State University, Murray, KY USA

B.Sc., Engineering Physics, May 2009

- Summa Cum Laude with Honors Diploma
- Honors Thesis: Development of a Reinforcement Learning Agent to Play Ms. Pac Man
- Minor in Mathematics and Statistics
- GPA: 3.87/4.0

ACADEMIC AND PROFESSIONAL EXPERIENCE

Chief Technology Officer, Co-Founder *2025 to Present*

Line

- Shipped the production AI sales agent to 10 venues across the US and Canada, where it has worked over 600 real world guest inquiries from March to September 2026, with over 80% of these conversations fully resolved by the sales agent.
- Designed a pre-send hallucination gate that checks every factual claim from the agent. This gate corrected 24.3% of all agent messages, removing agent fabrications before they reached a guest.
- Stood up an AI-first development pipeline — an orchestrator that runs isolated coding agents on dedicated infrastructure and returns a reviewed, quality-gated pull request with no human in the loop.
- Drove a two-engineer team to 1,624 merged pull requests and 9,533 commits in 14 months, with 75% agent-authored under human review.
- Built the test and review foundation that enabled our velocity: 20,000+ automated tests including 124 Playwright end-to-end journeys, AI review on every pull request, and sharded self-hosted CI that returns a full verdict in ~12 minutes and gated 726 production deploys.
- Partnered with the founding team on fundraising, technical diligence, and company strategy, translating architecture and technical risk for investors.

Director of Engineering, Codecov *2022 to 2025*

Sentry

- Devised and led Codecov's transition from a single-purpose code coverage tool to a robust code quality platform.
- Scaled the Engineering, Design, and Product functions across multiple product areas, hiring both individual contributors and managers.
- Successfully transitioned a team of 30 to the principles, culture, and working practices of Sentry with minimal attrition.
- Led coordination efforts between Sentry and Codecov teams to launch key product integrations and new features.
- Moved Codecov from closed source to a Fair Source License project, making all source code and product planning publicly available.

Chief Technology Officer, Co-Founder *2018 to 2022*

Codecov

- Helped scale our team from 3 to 30+ employees and contractors.
- Multiplied annual recurring revenue by 10x from 2019 - 2022.
- Worked alongside our Vice President of Engineering to shape Codecov's engineering processes and best practices.
- Directly led the solutions engineering, technical support, and product teams.

- Served as the technical point of contact for stakeholders from many of the world's largest companies.
- Worked directly with Codecov's CEO to set product roadmap and long-term technical direction.

Chief Technology Officer, Co-Founder *2012 to 2018*

GameWisp, Inc.

- Grew and managed a diverse team from 1 to 10 employees and contractors.
- Developed production quality software responsible for over $5M in Annual Recurring Revenue.
- Designed performant and cost-effective video upload, encoding, and delivery systems utilized by thousands of active users.
- Designed, developed, and maintained production quality REST and WebSocket APIs used by hundreds of third-party developers.
- Routinely developed features and products and shipped production-quality code.
- Successfully assisted GameWisp's CEO with multiple rounds of venture capital fund-raising.
- Performed wireframing, A/B testing, data analysis, and user interviews for GameWisp products.

Research Assistant *2009 to 2014*

Department of Electrical Engineering and Computer Science; Vanderbilt University

- Implemented internal development processes for all laboratory software projects using C++/Qt, Ruby on Rails, R, and Git.
- Independently designed, sourced, assembled, and flight tested unmanned aerial vehicles and balloons for archaeological field use in the Andes of South America.
- Supported archaeological survey teams on-site in the Peruvian highlands by providing hardware and software development.
- Provided a production quality implementation of the General Visualization Abstraction Algorithm in C++.

REFEREED PUBLICATIONS

Running a Red Light: An Investigation into Why Software Engineers (Occasionally) Ignore Coverage Checks. Alexander Sterk, Mairieli Wessel, Eli Hooten, Andy Zaidman. Proceedings of the International Conference on Automation of Software Test (AST), 2024. *Runner-up Paper Award.*

Of Drones and Deep Maps: Archaeological Placemaking with Aerial Robots, Photogrammetry, and GIS. Wernke, Steven A., Julie Adams, A., Eli Hooten, Gabriela Oré, Carla Hernández and Aurelio Rodríguez. THATCamp (The Humanities and Technology) conference, 2014.

Cancerous tweets: socially sharing sensitive health information. MD Anderson, JA Adams, ER Hooten. Systems, Man and Cybernetics (SMC), 2014 IEEE International Conference on.

Tablet Interaction Accuracy and Precision: Seated vs. Walking. ST Hayes, ER Hooten, JA Adams. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 2014. *Winner, Best Student Paper.*

Capturing Complexity: Toward an Integrated Low-Altitude Photogrammetry and Mobile Geographic Information System Archaeological Registry System. Steven A Wernke, Julie A Adams, Eli R Hooten. Advances in Archaeological Practice.

Communicative modalities for mobile device interaction. Eli R Hooten, Sean T Hayes, Julie A Adams. International Journal of Human-Computer Studies, 2013.

Comparing input error for mouse and touch input. Eli R Hooten, Julie A Adams. Systems, Man, and Cybernetics (SMC), 2011 IEEE International Conference on.

Visually-Cued Touch Gestures for Accurate Mobile Interaction. Sean T Hayes, Eli R Hooten, Julie A Adams. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 2011.

A Comparison of Communicative Modes for Map-Based Tasking. Eli R Hooten, Sean T Hayes, Julie A Adams. Proceedings of the Human Factors and Ergonomics Society Annual Meeting, 2011.

Multi-touch interaction for tasking robots. Sean Timothy Hayes, Eli R Hooten, Julie A Adams. Human-Robot Interaction (HRI), 2010 5th ACM/IEEE International Conference on.

PRESENTATIONS

Beyond the Basemap: Site and Landscape Survey through Low Altitude Aerial Photogrammetry and Mobile GIS in the Andes. Steven A. Wernke, Julie A. Adams, Eli Hooten, Gabriela Oré, Carla Hernández, Aurelio Rodríguez, and Giancarlo Marcone. MOBILIZING THE PAST FOR A DIGITAL FUTURE: The Potential of Digital Archaeology, 2015.

COMMITTEE APPOINTMENTS

Montessori School of Huntsville, Secretary, Board of Directors — 2023 to 2025

Delft University of Technology (Netherlands), Graduate Thesis Committee Member and Advisor for Alexander Sterk — 2022 to 2023

GameWisp, Inc., Board Member — 2016 to 2019

20th IEEE International Symposium on Robot and Human Interactive Communication, 2011 — Associate Editor

8th IEEE International Workshop on Safety, Security, and Rescue Robotics, 2010 — Program Committee Member

AWARDS AND HONORS

International Conference on Automation of Software Test (AST)

- Runner-up Paper Award, 2024 (as advisor to Alexander Sterk)

Human Factors and Ergonomics Society

- Winner, Best Student Paper, 2014

American Society for Engineering Education

- National Defense Science and Engineering Graduate Fellowship, 2009 – 2012

National Science Foundation

- Graduate Research Fellowship Honorable Mention, 2010
- Graduate Research Fellowship Honorable Mention, 2009

Murray State University

- Alumni Association Top 22 under 40, 2019
- Presidential Fellowship, 2004 – 2009
- Dean's List for Academic Excellence, 2004 – 2008
- Undergraduate Research and Scholarly Activity Grant, 2009
- Honors Program Outstanding Senior Graduate, 2009
- Engineering and Physics Outstanding Senior Graduate, 2009
- Starick Scholarship Recipient, 2008
- Student Organization President of the Year Nominee, 2008
- Honor Society Inductions: Society of Automotive Engineers, Sigma Pi Sigma Physics Honors Society, Pi Mu Epsilon Mathematics Honors Society, Omicron Delta Kappa Honors Society (all 2007)

SERVICE

Montessori School of Huntsville
Huntsville, AL USA

Secretary, Board of Directors, 2023 – 2025

- Coordinated fundraising, service, and charitable work for the school alongside the board.
- Led the consolidation of the school from two campuses to one, conducting a full impact study to support the decision.
- Helped coordinate the subsequent expansion of the single campus to accommodate faculty, students, and staff.

Honors Program
Murray State University, Murray, KY USA

President of Student Council, 2007 – 2008

- Successfully proposed the addition of an Honors Math course to the Honors Program curriculum.
- Oversaw council activities across all philanthropy, outreach, academic, social and service activities.
- Acted as a liaison between the Honors Program and all other campus organizations.

Philanthropy Committee Chair, 2007 – 2009

- Oversaw the planning and execution of all philanthropic activities.
- Raised thousands of dollars for the Waterfield Library annually.
- Personally participated in all program fund raising events.

Department of Engineering and Physics
Murray State University, Murray, KY USA

Moonbuggy Team Co-Captain, 2007 – 2009

- Participated in NASA's annual Great Moonbuggy Race.
- Worked with a team to design a race vehicle from scratch.
- Winner of 2007 Unique Design Award
- 2008 Second Place Finish
- 2009 Fourth Place Finish

Baja SAE Team Co-Captain, 2007 – 2008

- Designed with a student team MSU's inaugural Baja car: a single seat, motorized, off-road vehicle.
- Personally designed and fabricated the vehicle's suspension system.
- Responsible for acquiring and managing corporate sponsorships.

REFERENCES

References available upon request.